

AIMO Proof Pilot — Final Grade

Problem (#_ID): 5_POLYGO

Submission: model_8

Scoring scale (0/1/6/7) — two binary decisions:

Step 1. Is the solution essentially complete?

- If **no**: award **1** for significant partial progress, otherwise **0**.
- If **yes**: award **7**, or **6** if there is a minor error or omission.

Grade against the **markscheme** (docs/markschemes.pdf).

1. Final mark

Final mark (choose one):

2. Reasoning

Justify the mark with explicit reference to the markscheme points (e.g. ME1) where possible.

Correct solution with converse checked correctly. A different approach to the mark scheme by the Rational Root Theorem and factorising, then case analysis on the coefficients, but it works well. It does make some assertions about multiple roots which are not correct in the $A = 1$ case; however, this is not needed for the argument and so does not affect the conclusion. We therefore award **7**.